

39° below the horizontal

- 2 (a) The linear momentum of a body is the product of the mass and its velocity. The linear momentum is in the same direction as its velocity.

- 2 (b) (i) By C.O.M.
$$mu = m_1v_1 + m_2v_2 \rightarrow (800)(165) = (170)(v_1) + (630)(235)$$
$$v_1 = -94.4 \text{ m s}^{-1}$$
$$\text{magnitude of } v_1 = 94.4 \text{ m s}^{-1}$$
$$\text{direction: in the opposite horizontal direction as larger fragment (or to the left, or backward)}$$

- (ii) Initial KE = $\frac{1}{2} (800) (165)^2 = 10.890 \text{ MJ}$
Final KE = $\frac{1}{2} (170) (94.4)^2 + \frac{1}{2} (630) (235)^2 = 18.153 \text{ MJ}$
Energy released by explosion = final KE – initial KE
 $= 7.26 \times 10^6 \text{ J}$